QuizRx Frontend

UML Documentation of the Whole Application

Repository	quizrx-frontend
Stack	Next.js 15 (App Router), React 19, TypeScript, Apollo Client (GraphQL), Zustand, Firebase Auth, Tailwind CSS / Radix UI
Hosting	Docker image on Google Cloud Run, built by Cloud Build
Diagrams	24 (architecture, class, sequence, state, deployment)
Generated	2 November 2026

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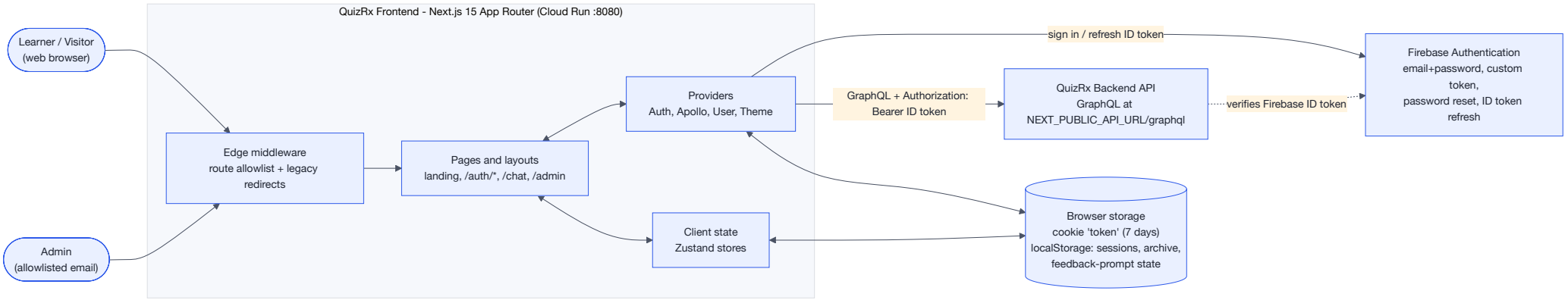
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- Notation: diagrams are written in Mermaid. Use case, component and deployment views use flowchart syntax with UML conventions (stadium shapes = actors / use cases, dashed arrows = include / dependency).
- Scope: everything reachable from routes allowed by **src/middleware.ts** is shown in detail. Modules that exist in the code but are not routed (concepts, graph, avatars, cms, subscription, legacy chat) are summarised separately.
- Server-side behaviour (backend API, Firebase) is shown only as seen from the frontend.

1. ARCHITECTURE

System Context

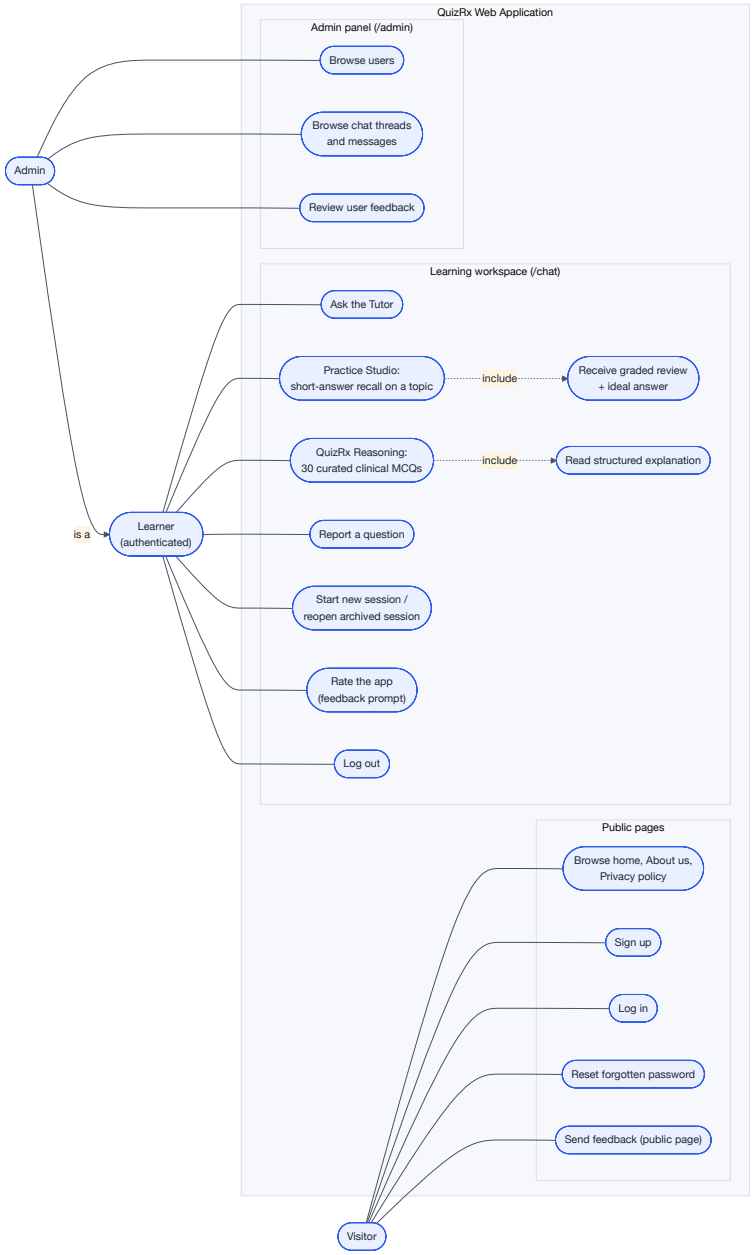
The QuizRx frontend is a Next.js 15 / React 19 app. Users authenticate with Firebase; every data call goes to the QuizRx backend over GraphQL with the Firebase ID token as a Bearer token. Client state lives in Zustand stores (partly persisted to localStorage) and React context.



1. ARCHITECTURE

Use Case Diagram

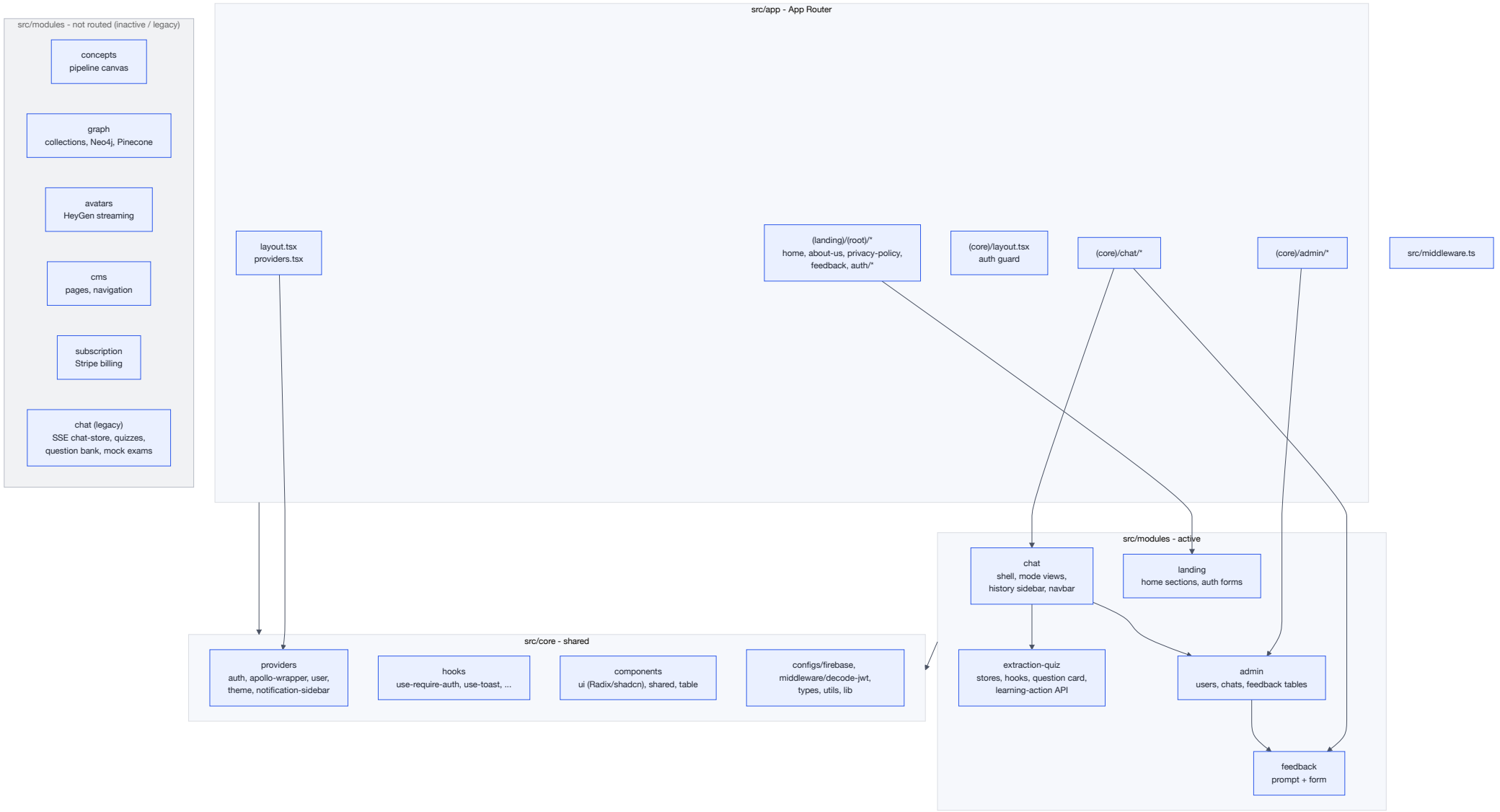
Actors and the goals they can achieve in the currently routed application. An Admin is a Learner whose email is on the client-side admin allowlist.



1. ARCHITECTURE

Package / Component Diagram

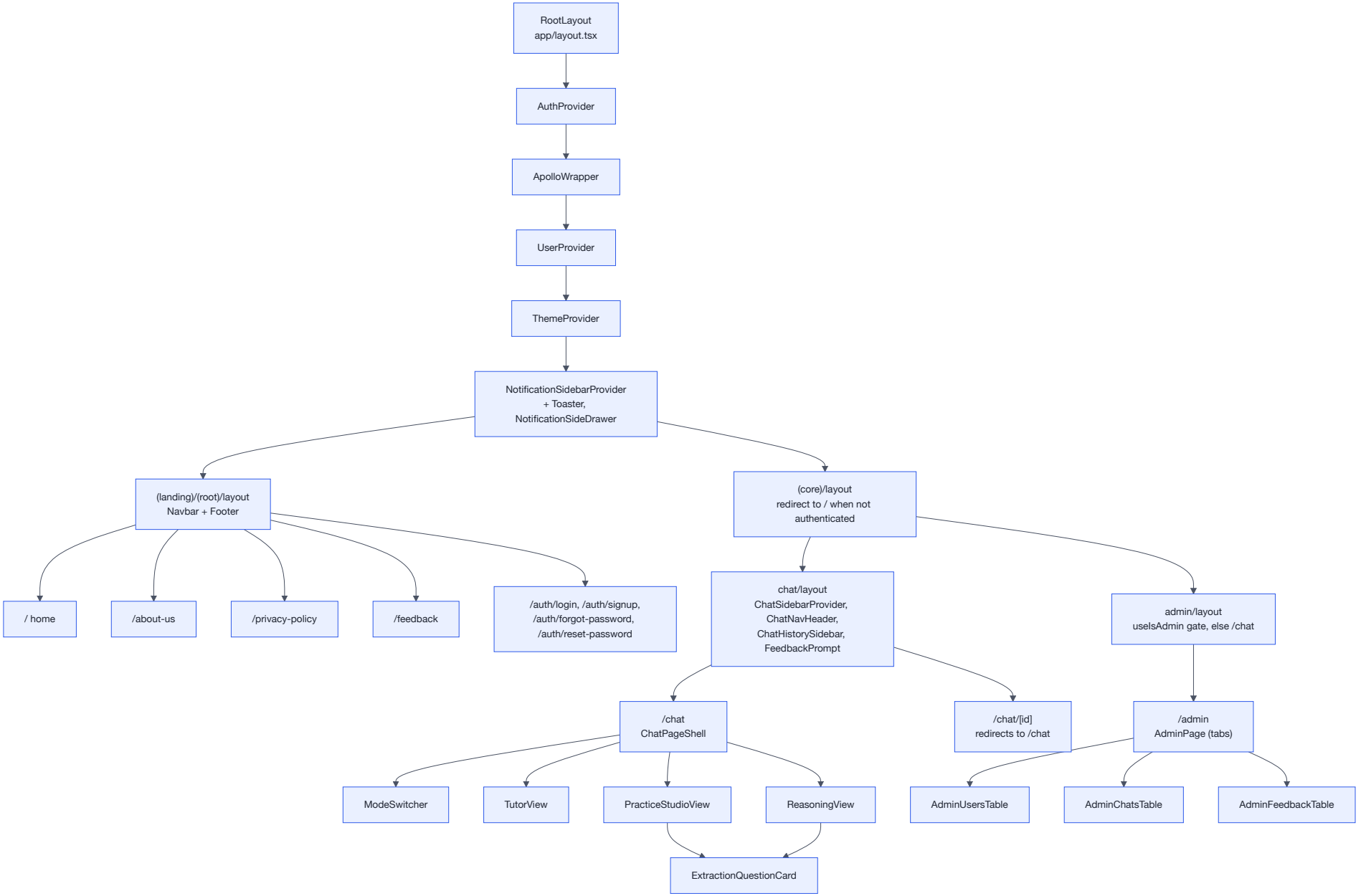
Source layout under src/. Arrows show compile-time dependencies. Modules in the grey box are present in the codebase but are not reachable from any route that the middleware currently allows.



1. ARCHITECTURE

Route Tree and Provider Composition

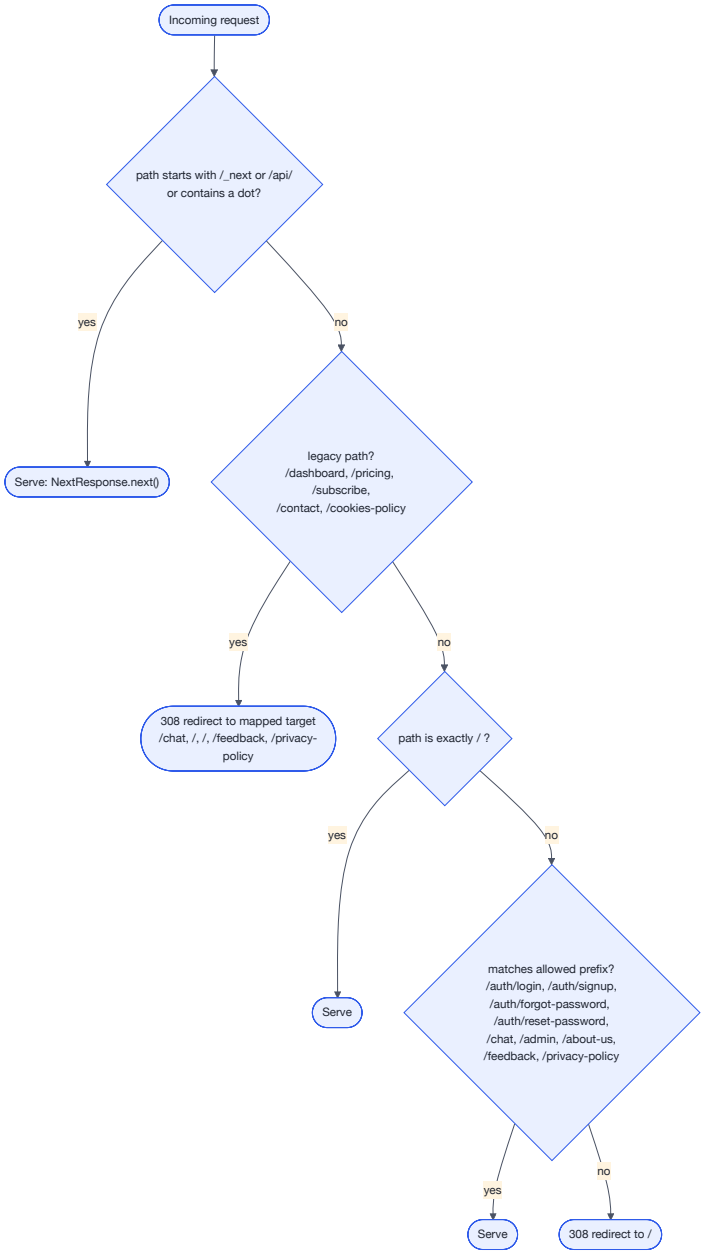
How the root layout composes global providers and which layouts, pages and top-level components render for each route.



1. ARCHITECTURE

Activity Diagram - Edge Middleware Routing

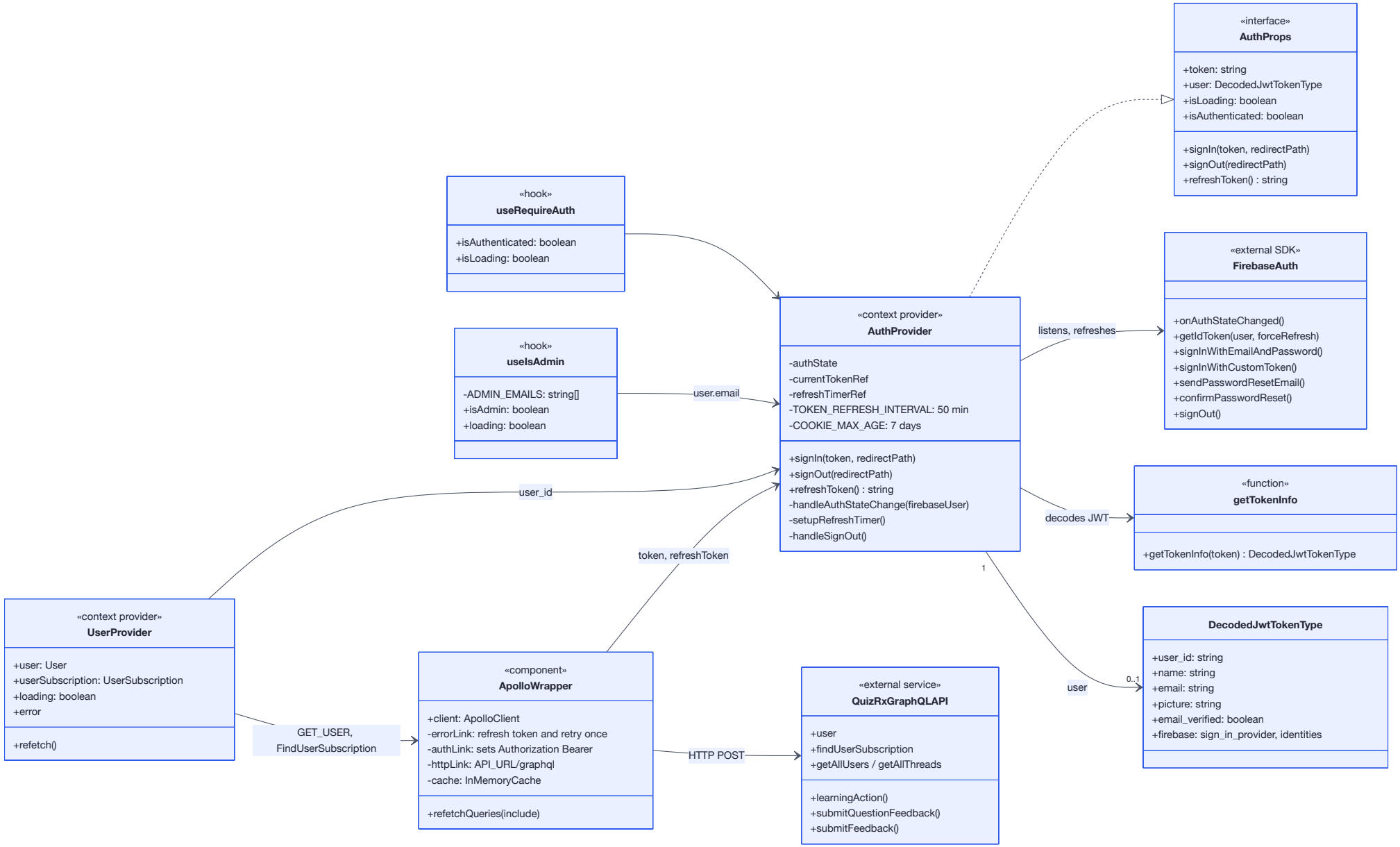
Every page request passes through src/middleware.ts. Only an allowlist of routes is served; legacy URLs are permanently redirected.



2. CLASS DIAGRAMS

Authentication and Data Access (src/core)

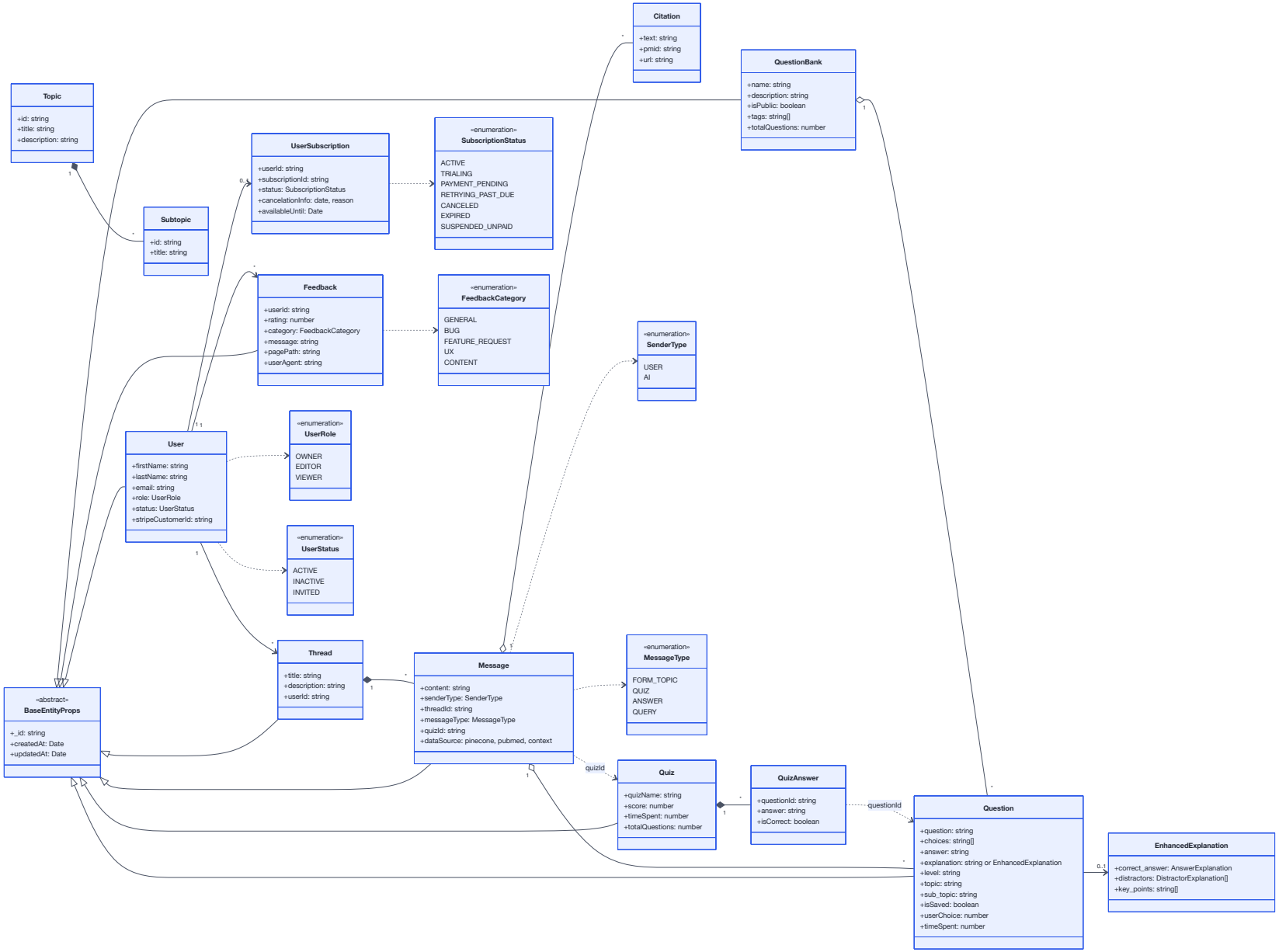
AuthProvider owns the Firebase session and the 'token' cookie. ApolloWrapper builds the GraphQL client (errorLink, authLink, httpLink) and silently refreshes expired tokens once. UserProvider loads the current user and subscription.



2. CLASS DIAGRAMS

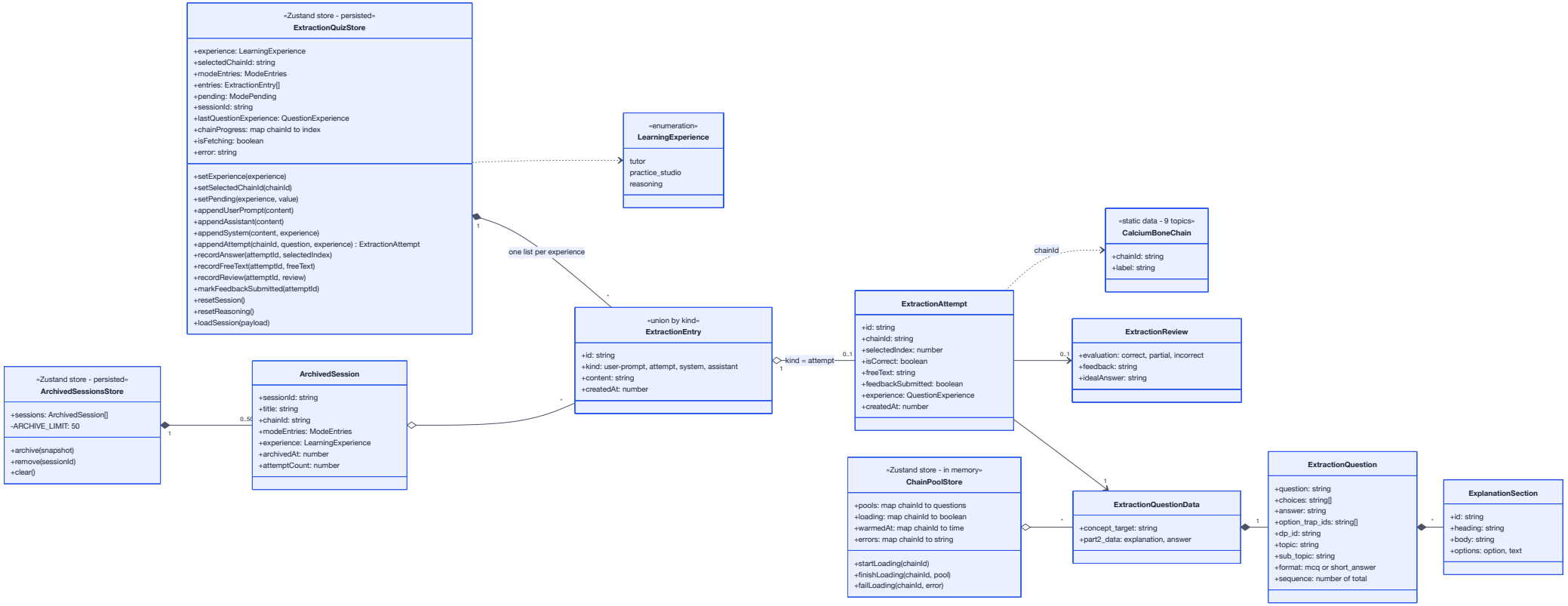
Domain Model (entities exchanged with the backend)

Entity types shared across modules. All persistent entities carry _id, createdAt and updatedAt (BaseEntityProps). Thread/Message/Quiz/QuestionBank belong to the legacy chat flow but are still read by the admin panel.



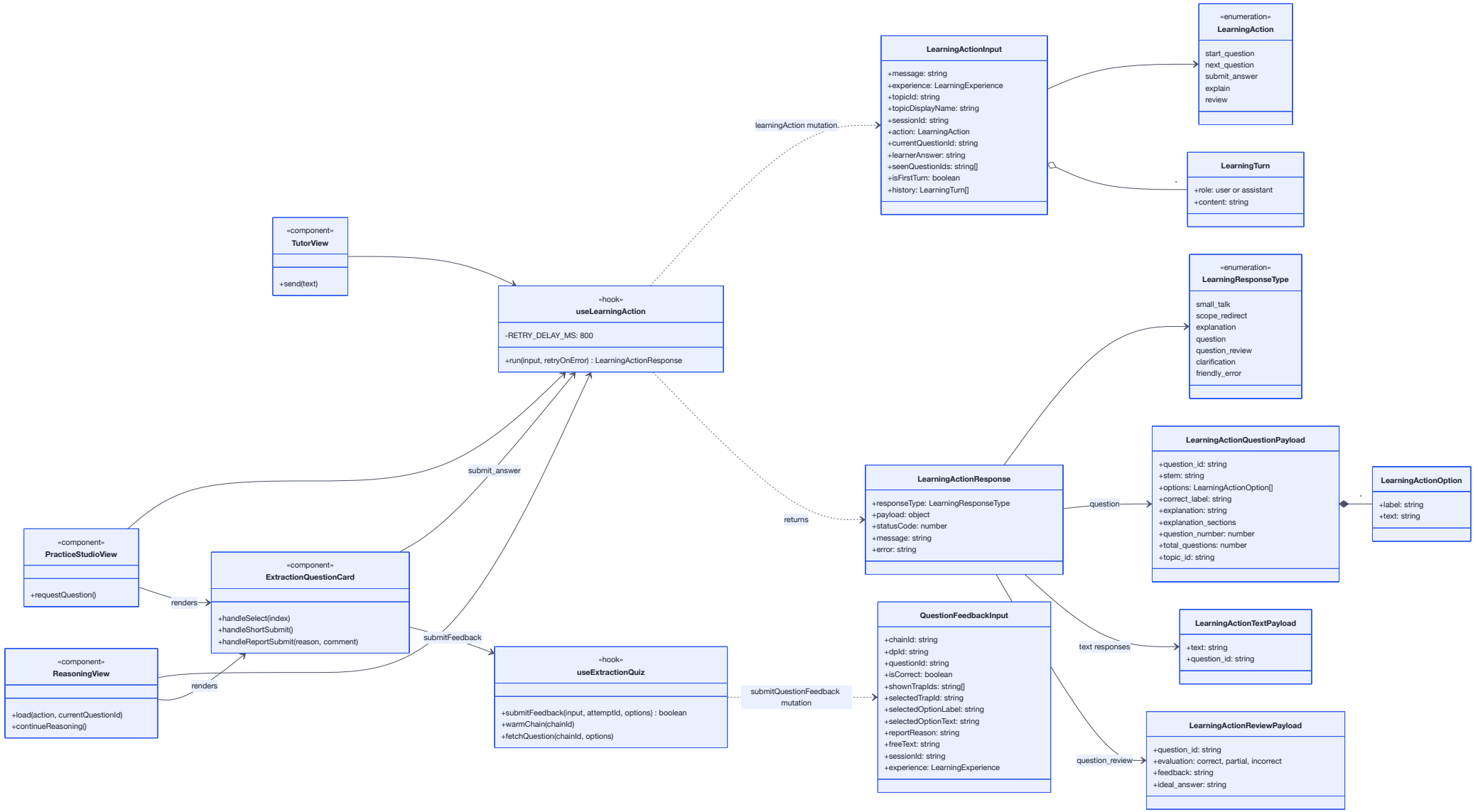
Learning Workspace - Session State (extraction-quiz stores)

The /chat page is driven by three Zustand stores. ExtractionQuizStore keeps one thread per learning experience (Tutor, Practice Studio, Reasoning) so switching modes never loses work. ArchivedSessionsStore keeps up to 50 past sessions. ChainPoolStore is an in-memory cache of authored questions per topic chain.



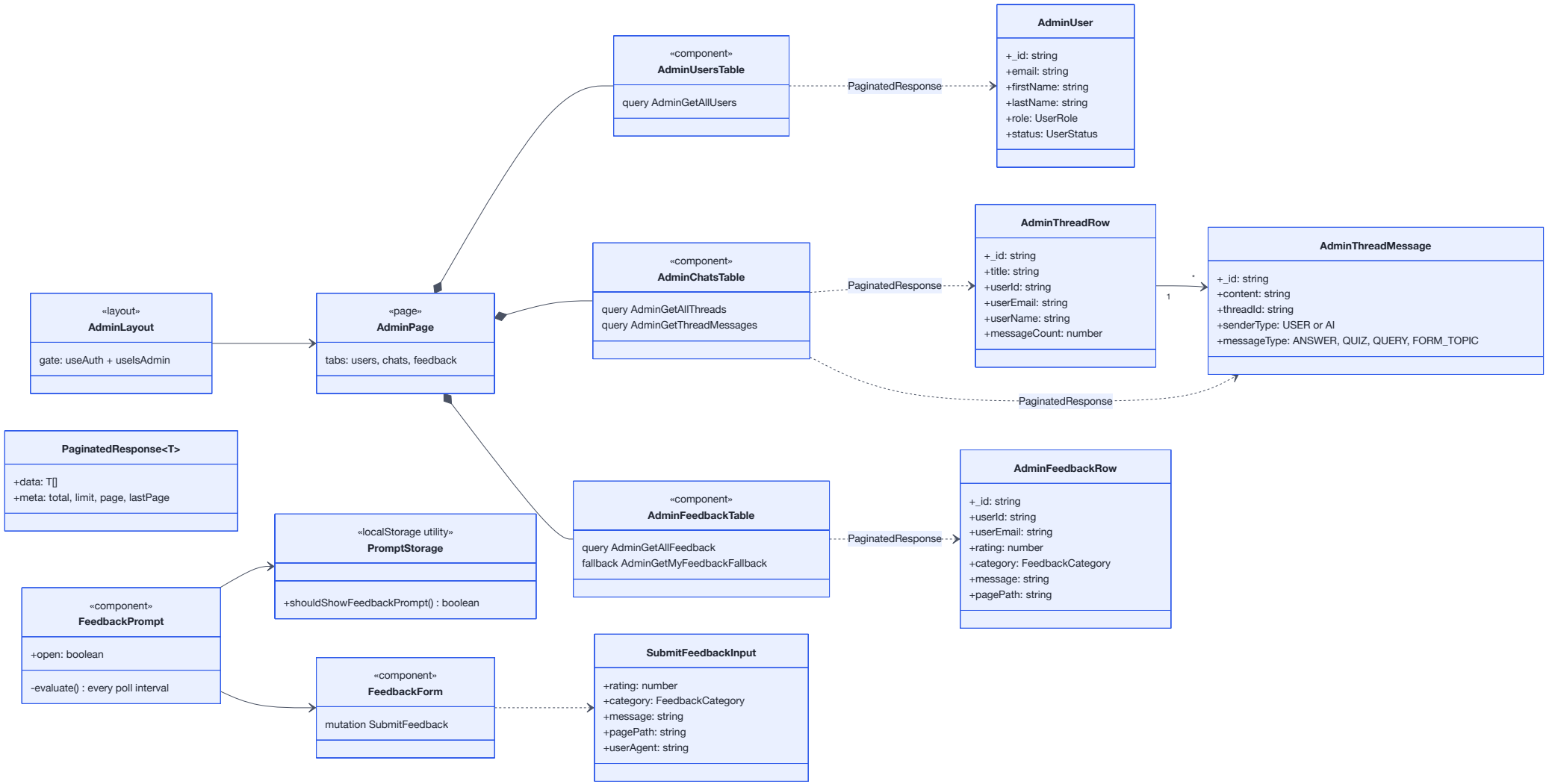
Learning Workspace - Learning Action API and UI

All three modes talk to a single GraphQL mutation, learningAction. The UI sends an explicit action (start/next question, submit answer) and switches on responseType. Answer analytics and question reports go through submitQuestionFeedback.



Admin Panel and Feedback

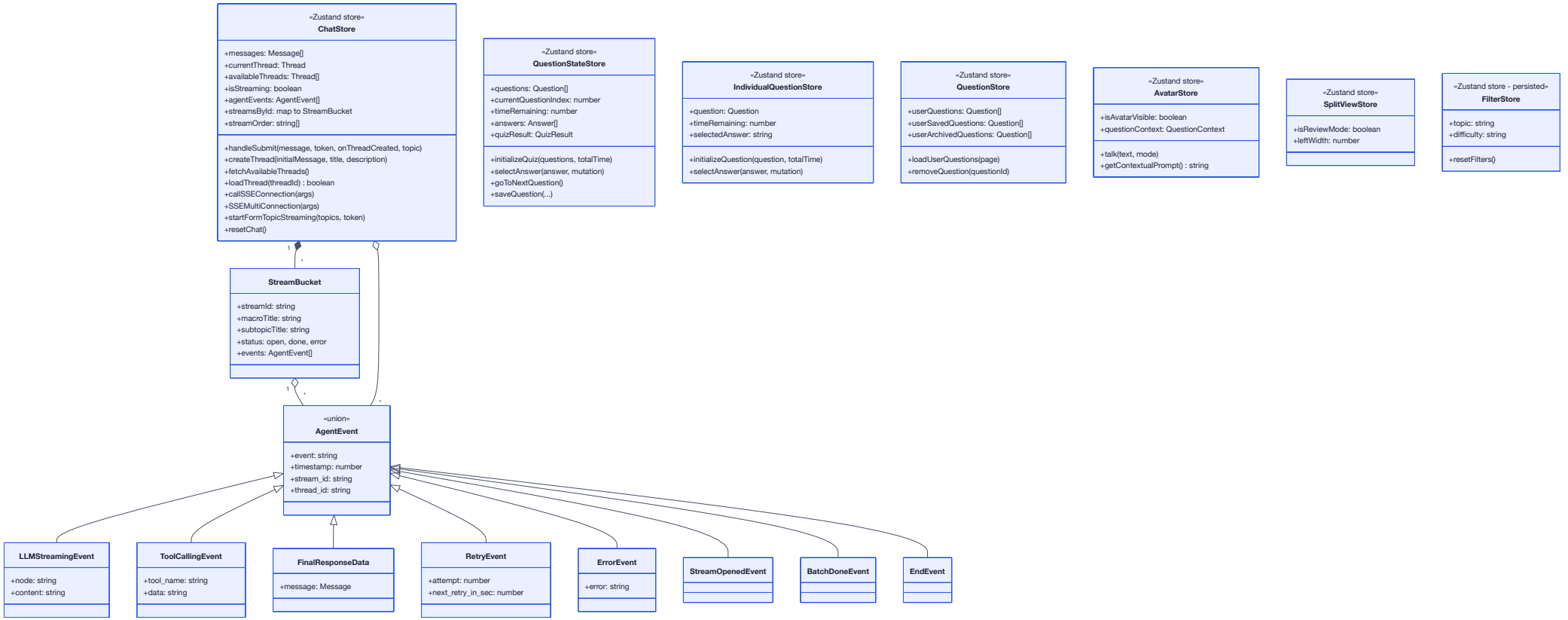
The admin panel is three read-only paginated tables. Access is gated client-side by useIsAdmin (email allowlist). The feedback prompt lives in the chat layout and decides when to appear from localStorage state.



2. CLASS DIAGRAMS

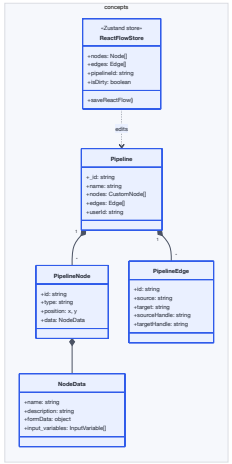
Legacy Chat Module (not rendered by the current /chat page)

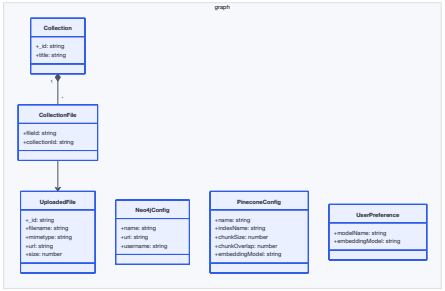
Stores from the previous chat experience: SSE streaming chat with agent events, quizzes, question banks and an avatar overlay. They are still in src/modules/chat/store but the current ChatPageShell does not use them.

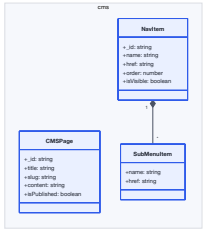


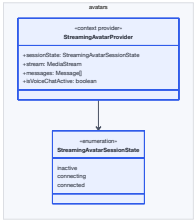
Inactive Modules - Concepts, Graph, CMS, Avatars

Types from modules that are in the repository but not reachable from any allowed route (pipeline builder, knowledge-graph data configuration, CMS, HeyGen streaming avatar).



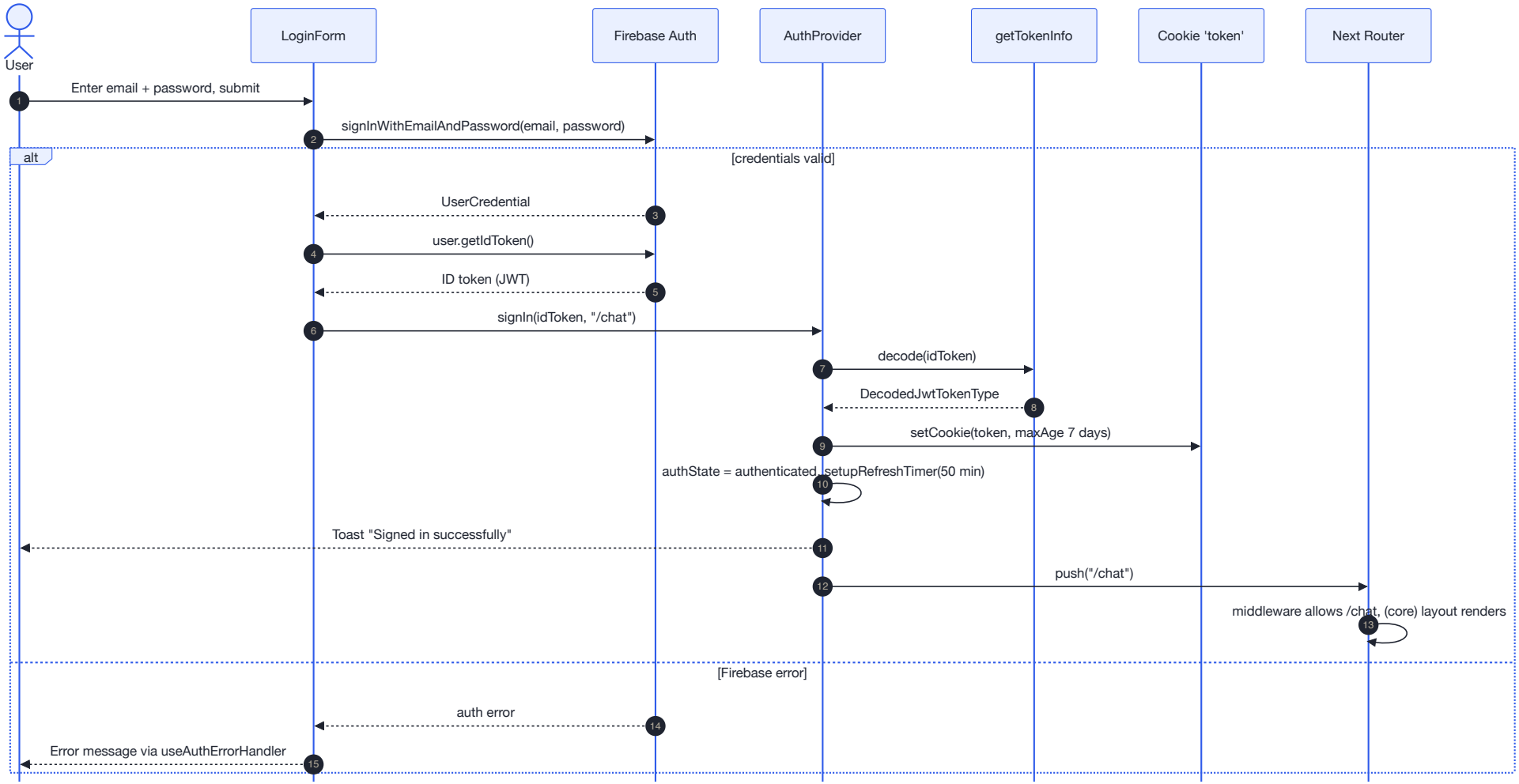






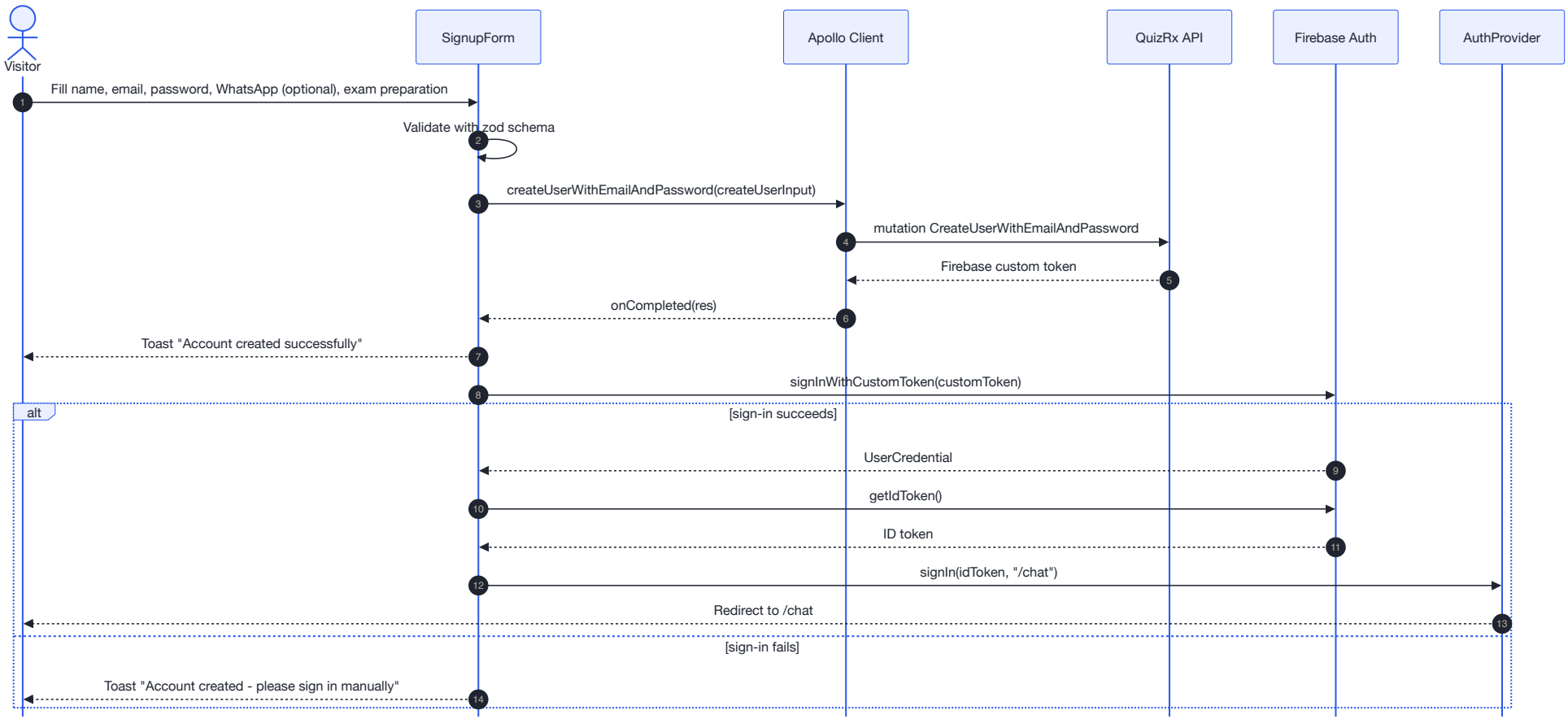
Log In with Email and Password

The login form authenticates against Firebase, then hands the ID token to AuthProvider, which validates it, stores it in a cookie and starts the 50-minute refresh timer.



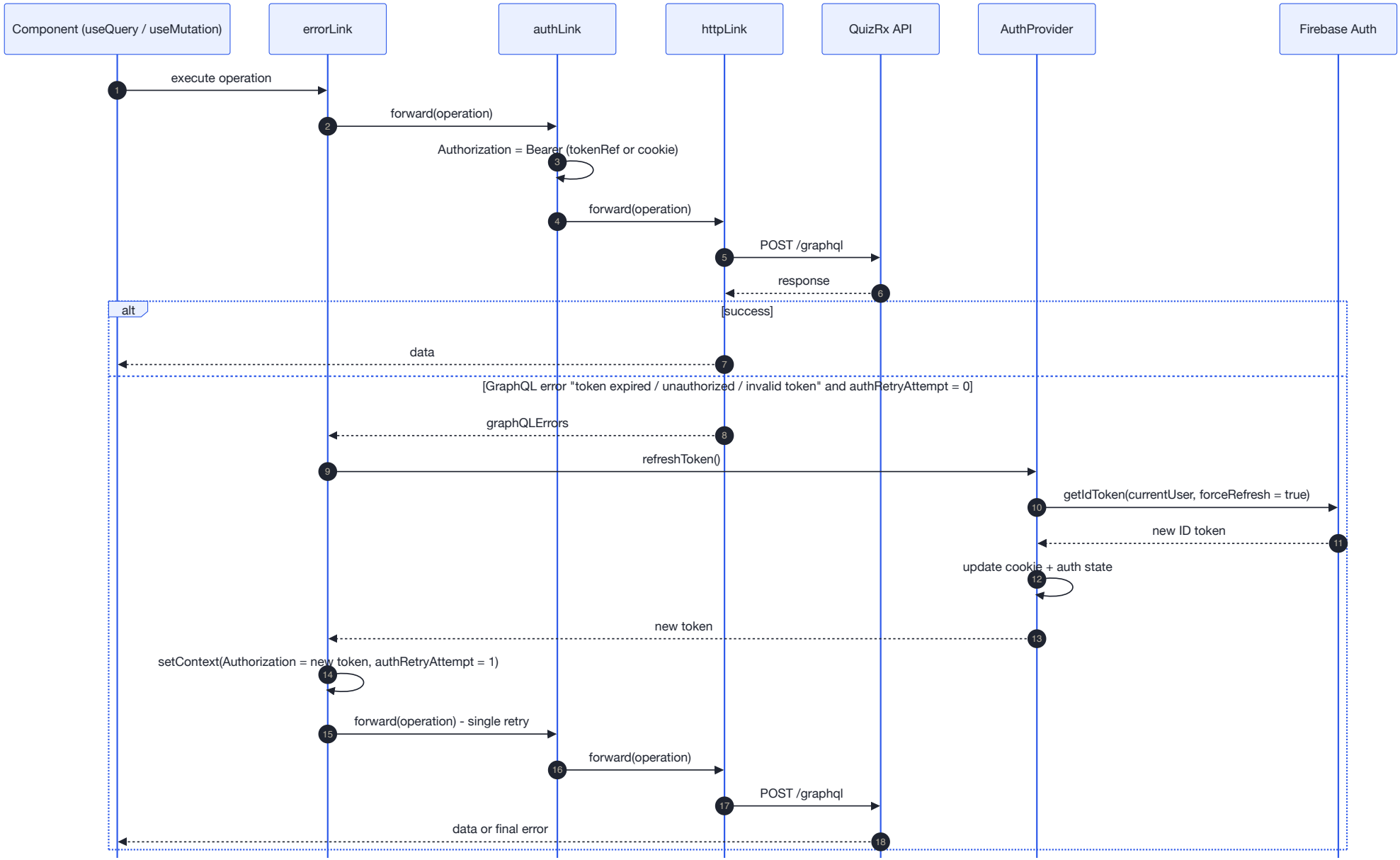
Sign Up

Account creation happens on the backend, which returns a Firebase custom token. The client exchanges it for an ID token and signs in.



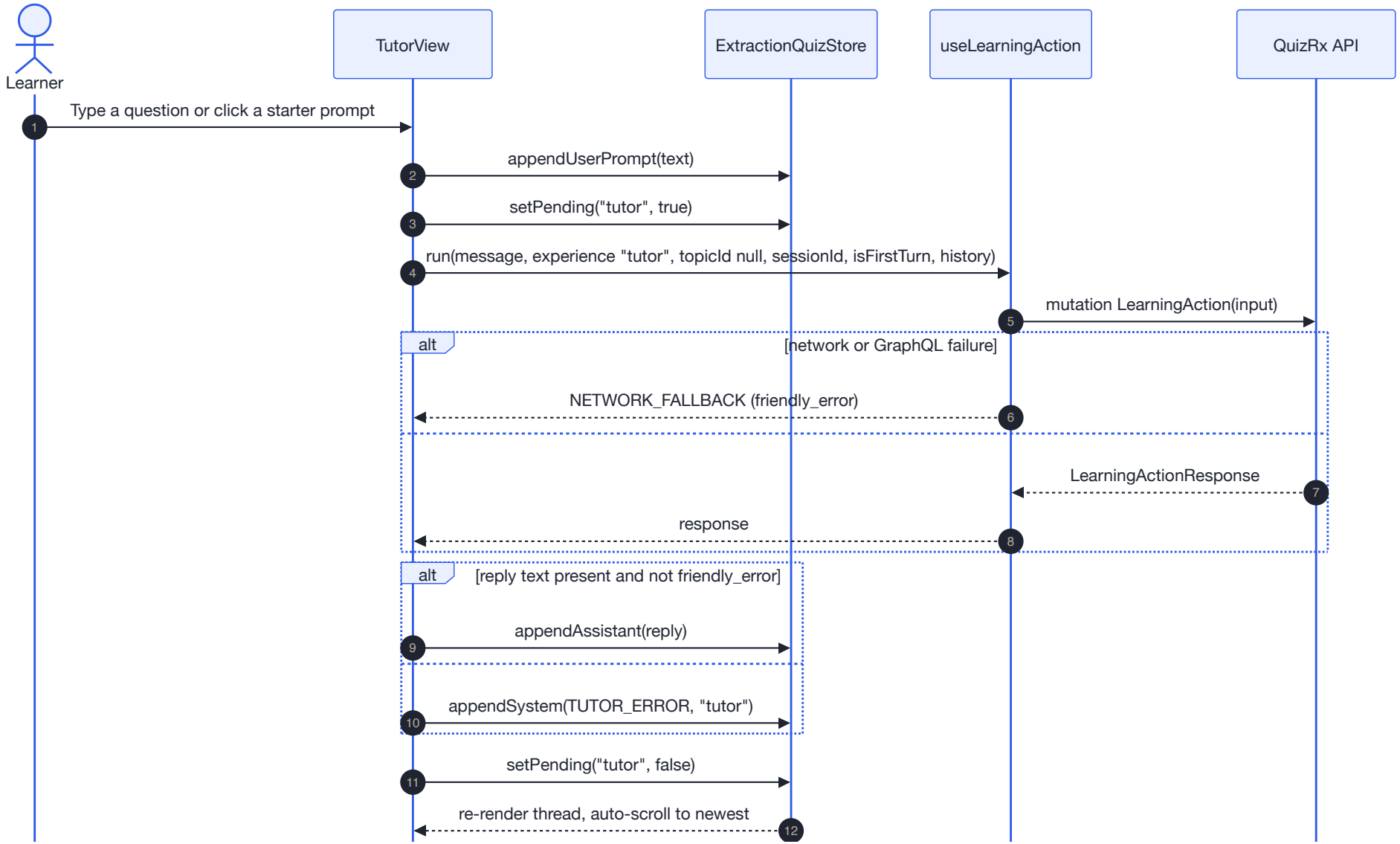
Authenticated GraphQL Request with Silent Token Refresh

Every GraphQL operation passes through errorLink, authLink and httpLink. If the backend reports an expired or invalid token, errorLink refreshes the Firebase ID token and retries the operation once.



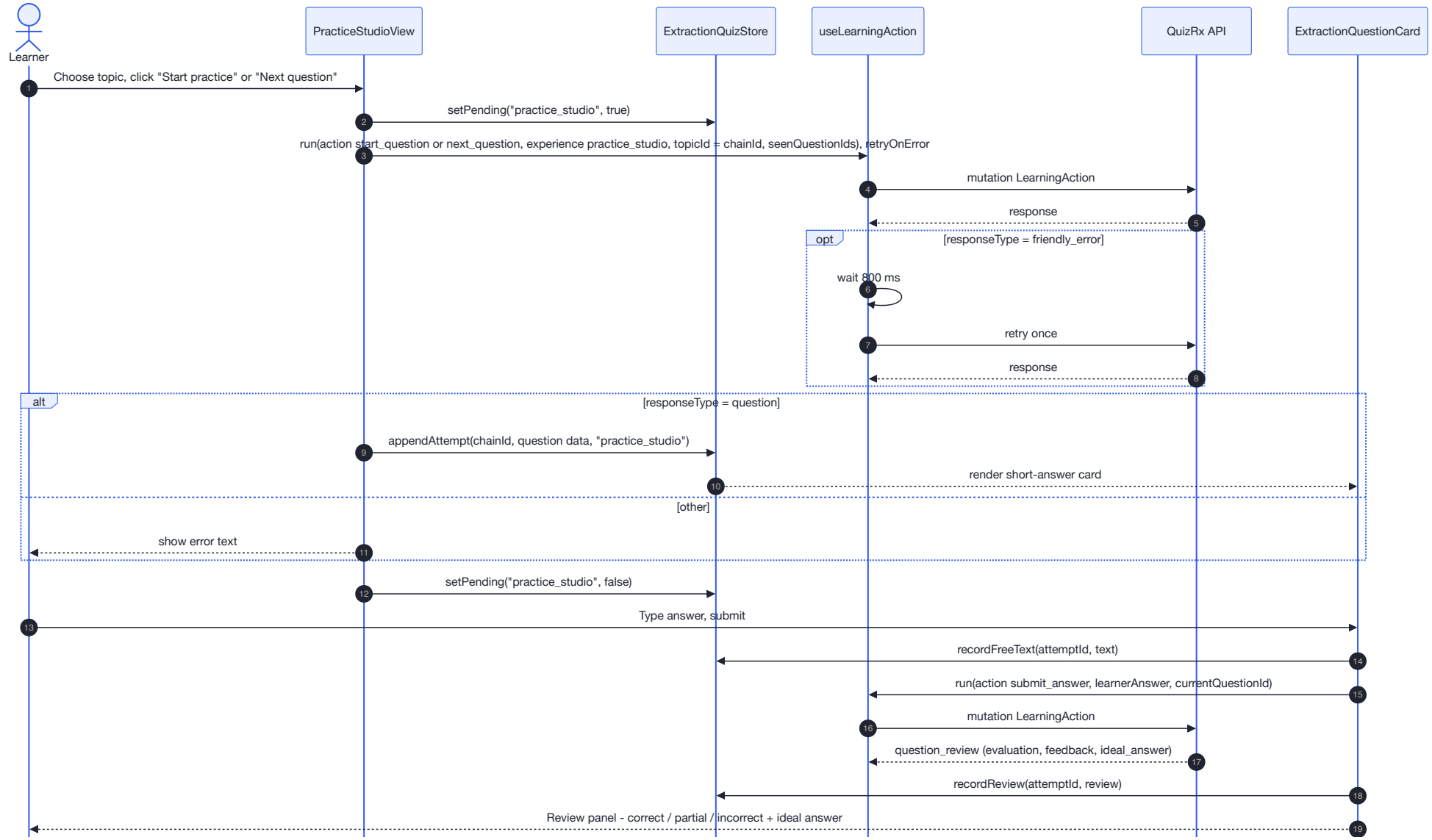
Tutor Mode - Ask a Question

Tutor is conversational. Prior turns are sent as history so follow-up questions keep context. Tutor never sends a topic.



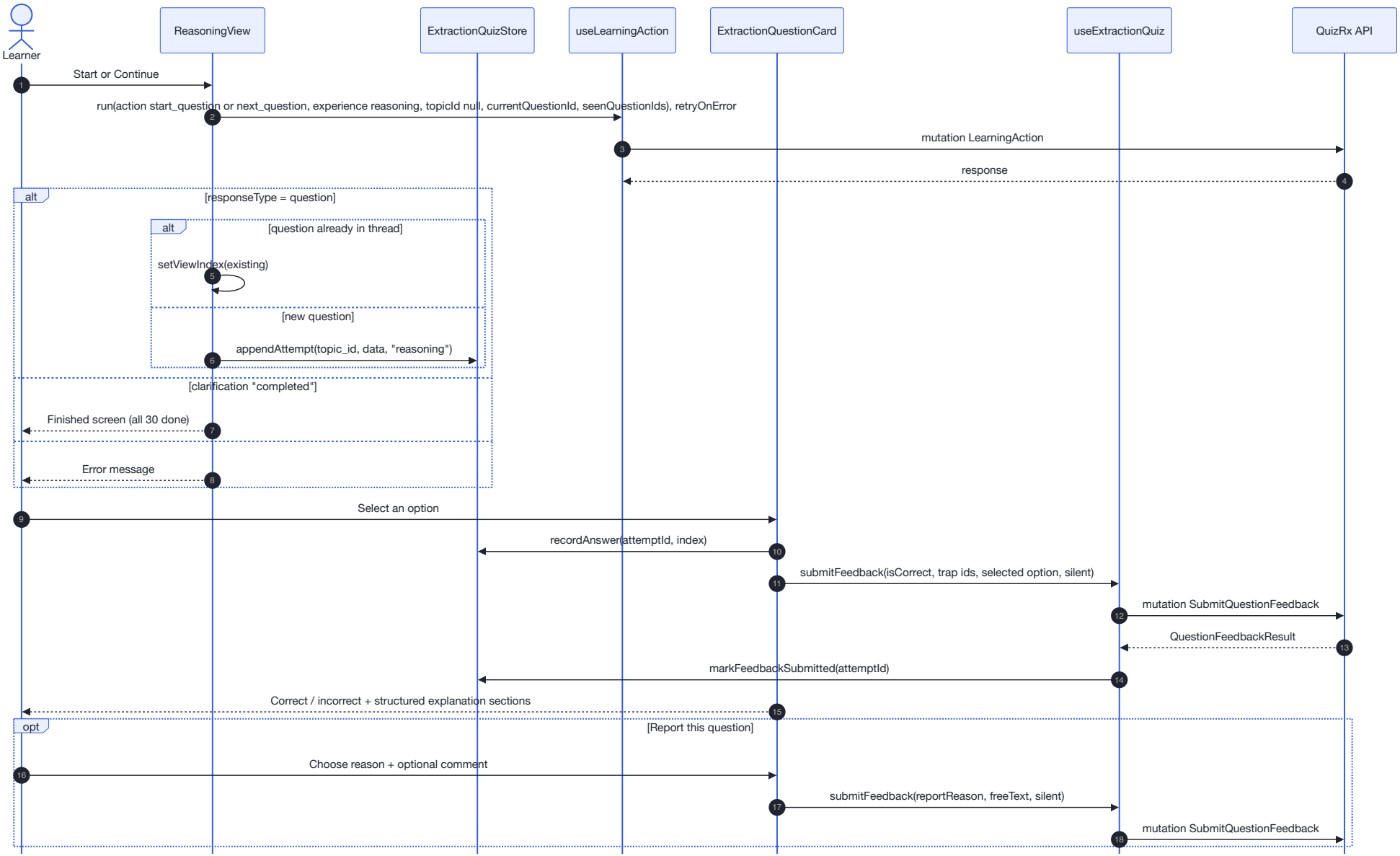
Practice Studio - Short-Answer Question and Graded Review

The learner picks a topic; the backend serves an unseen question; the typed answer is graded semantically and returned as a question_review.



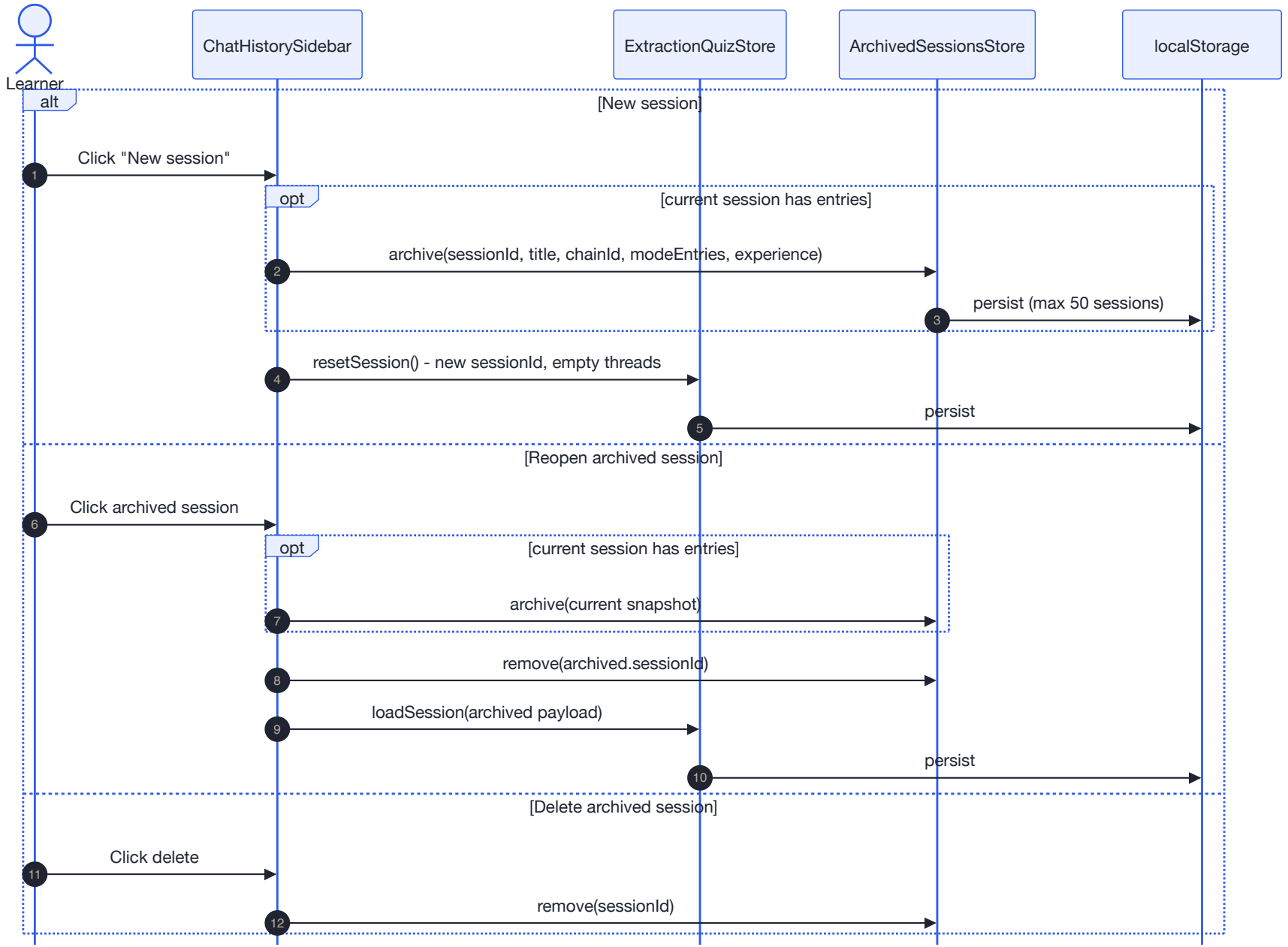
QuizRx Reasoning - Answer a Curated MCQ and Report

Reasoning walks a fixed sequence of 30 curated questions. Selecting an option records the answer and silently sends answer analytics; the learner can also report a question.



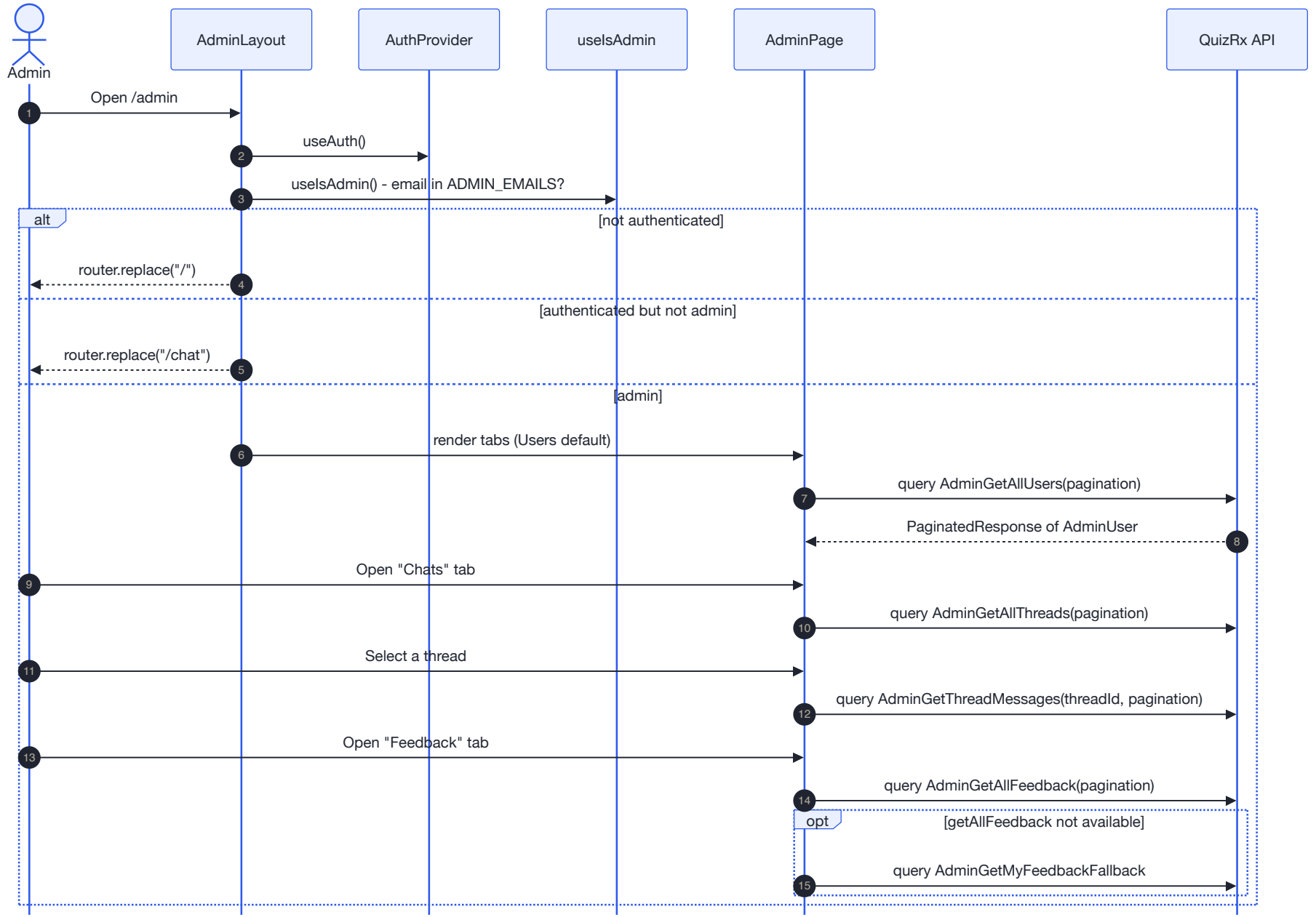
Session Management (Chat History Sidebar)

Sessions are kept on the device. Starting a new session archives the current one; reopening an archived session swaps it back in.



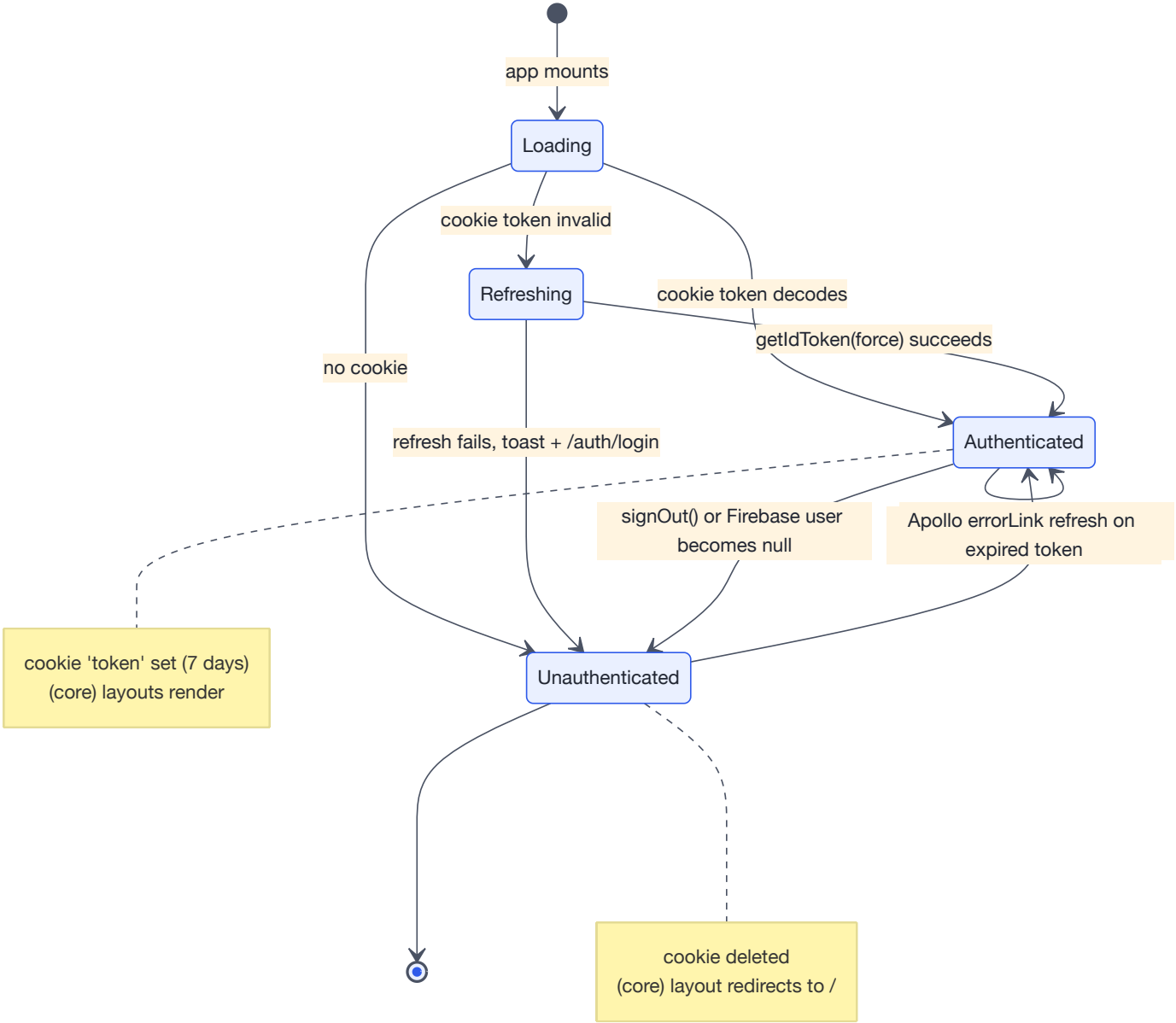
Admin Panel

The admin layout gates access client-side, then each tab loads its own paginated GraphQL query.



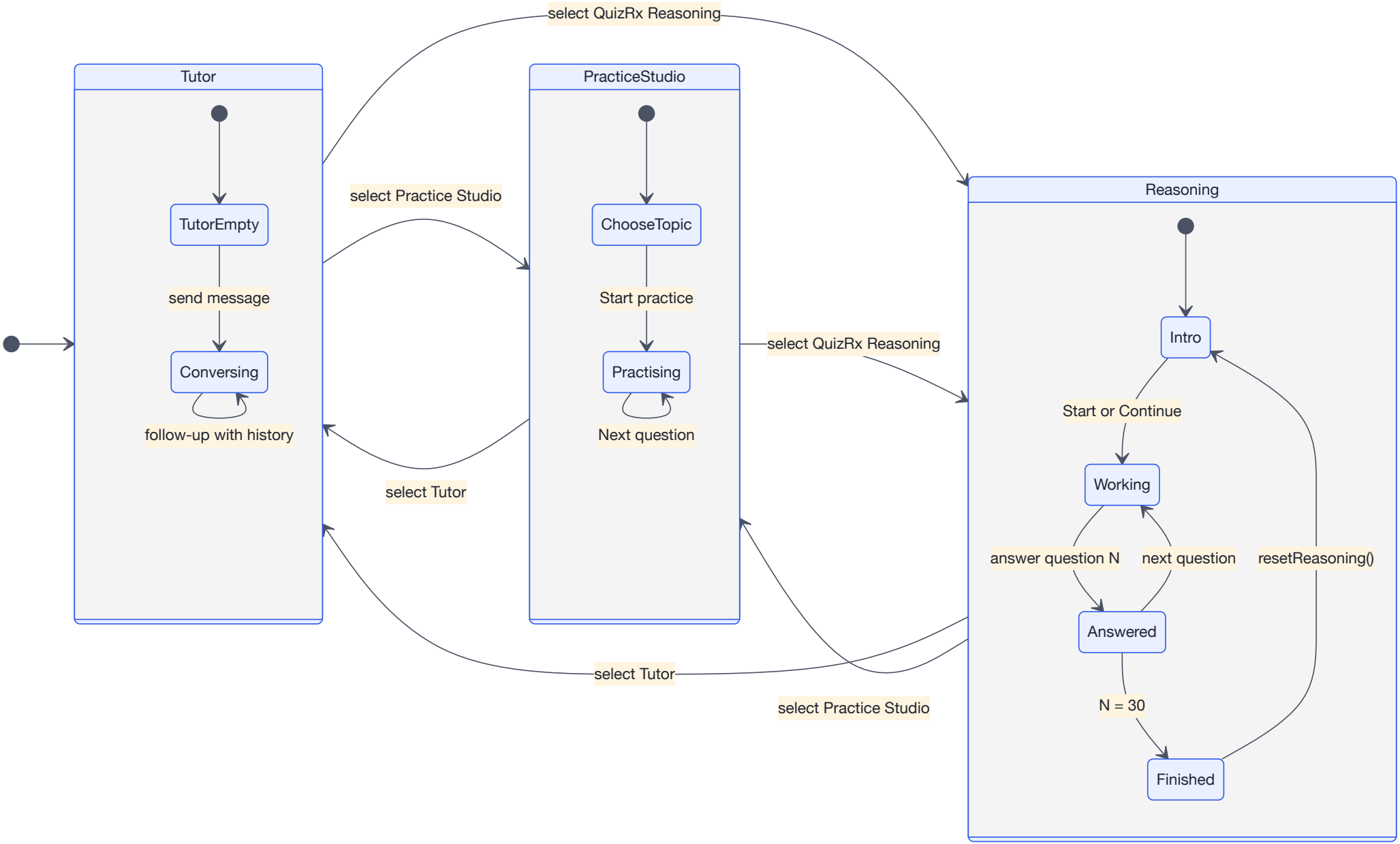
Authentication State (AuthProvider)

Lifecycle of the client-side auth state, including restore from cookie, periodic refresh and sign-out.



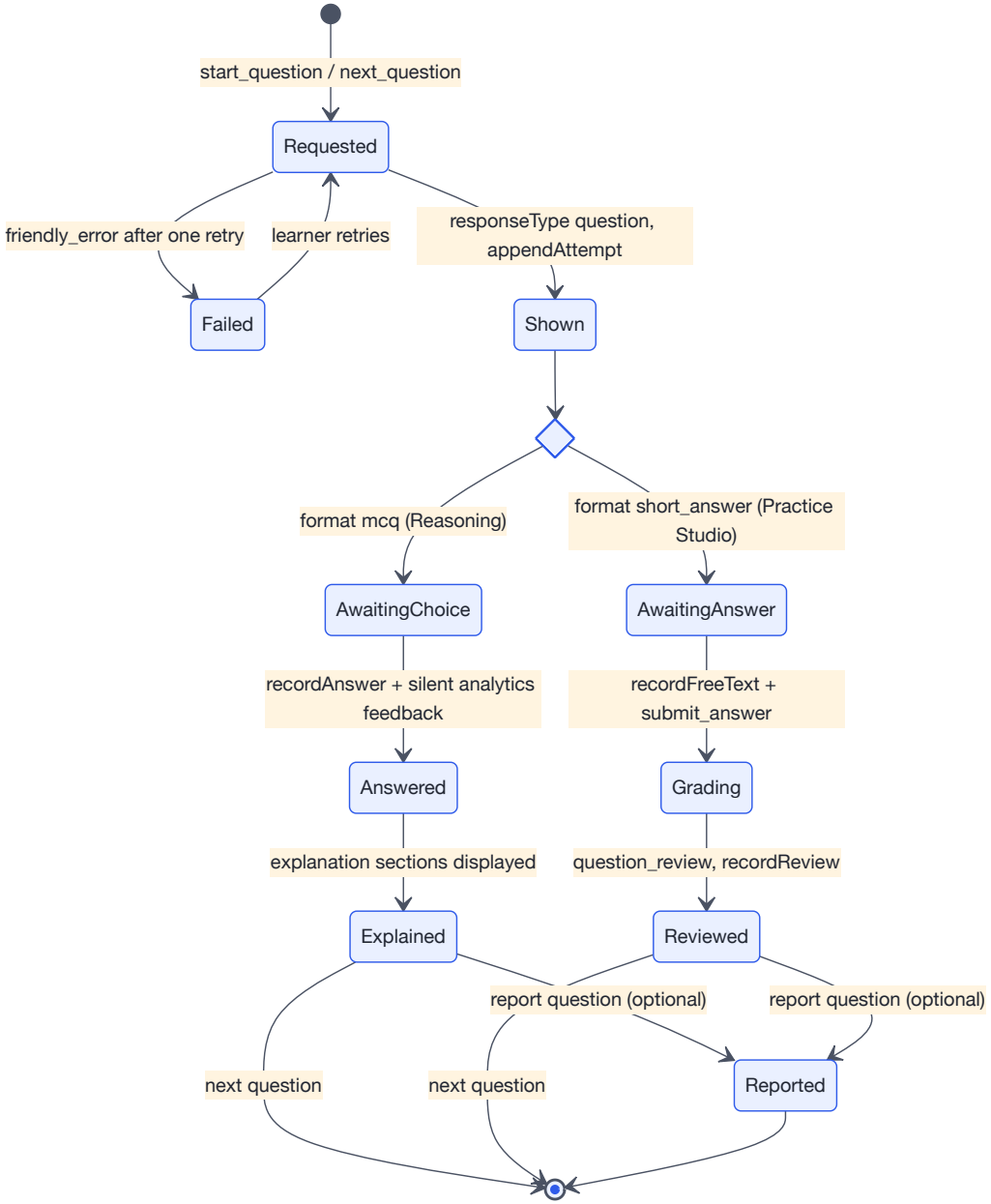
Learning Mode Switching

The ModeSwitcher changes ExtractionQuizStore.experience. All three views stay mounted, so drafts, open questions and scroll positions survive switching. Every visit starts in Tutor.



Question Attempt Lifecycle

States an ExtractionAttempt goes through. Reasoning uses multiple choice; Practice Studio uses short answers graded by the backend.



5. DEPLOYMENT

Deployment Diagram

Production builds run on Google Cloud Build: NEXT_* secrets are pulled from Secret Manager into .env, a multi-stage Docker image is built and pushed, and the service is deployed to Cloud Run.

